

Computational Appraisal of Gender Representativeness in Popular Movies – Paper Presentation

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What are they looking at and why?

- How under- or over-represented are women on-screen in popular movies?
- Can computational methods (face and gender detection) measure this representation more efficiently and accurately over time?

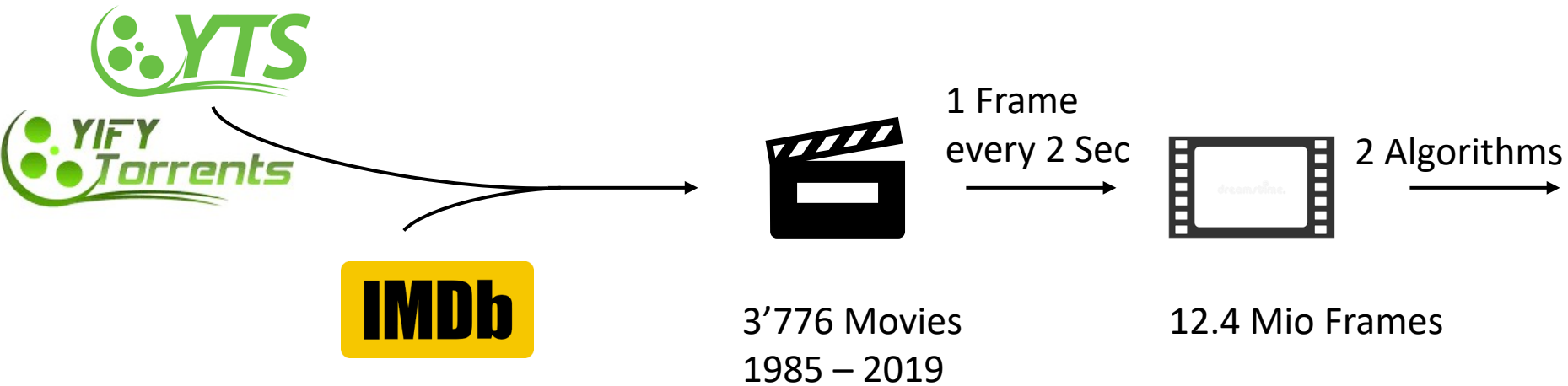
Computational appraisal of gender representativeness in popular movies

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How did they answer their questions?

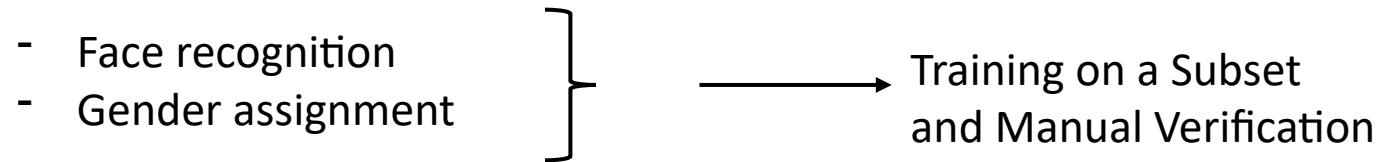


<https://de.dreamstime.com/flat-raster-film-frame-icon-filmrahmenraster-symbol-piktogramm-f%C3%BCr-den-flachbildfilm-auf-wei%C3%9Fem-grund-image16623082>

<https://en.wikipedia.org/wiki/YIFY>
<https://www.facebook.com/YIFYTORRENTS/>
<https://www.shutterstock.com/search/face-icon-front>
<https://uxwing.com/face-icon/>

How did they answer their questions?

2 Algorithms



- How good are their algorithms
 - 92% accuracy for face recognition
 - 73.9% accuracy for gender assignment
 - 65% for female faces correctly assigned
 - 84.5% for male faces correctly assigned

How did they answer their questions?

2 Algorithms

- Face recognition
- Gender assignment

Haar-like features can help to identify faces.

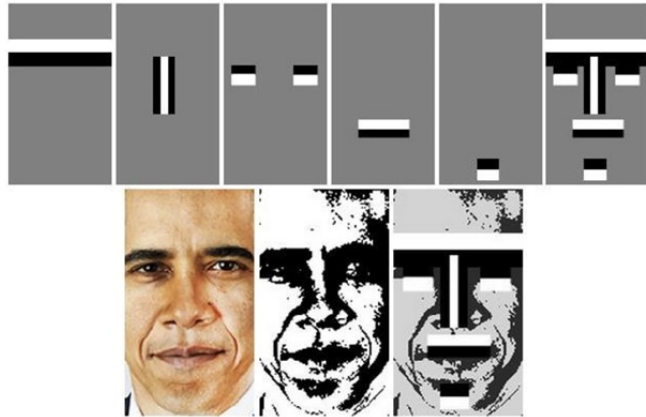


Figure: Kadir et al. (2014): an application of Haar-like features²

```
# Where could all these faces be?

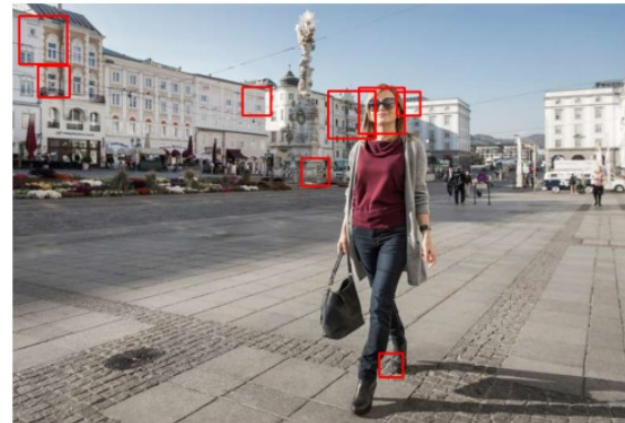
# Let's draw rectangles around the faces and plot the image
for (x, y, w, h) in faces_finer:
    cv2.rectangle(people_image, (x, y), (x+w, y+h), (255, 0, 0), 2)

# Display the image with the rectangles drawn around the faces
plt.imshow(people_image)
plt.title('Face Detection')
plt.axis('off')
plt.show()

# The CascadeClassifier operates on different scales and looks for features that resemble face features such as eyes, nose, and the mouth region
# Sometimes it incorrectly classifies objects as faces -- if they have similar Haar-like features as faces...
```



Face Detection



How did they answer their questions?

2 Algorithms

- Face recognition
- Gender assignment

```
# Plot one random example per ethnicity
fig, axes = plt.subplots(1, len(unique_races), figsize=(15, 3))

for idx, race in enumerate(unique_races):
    # Get first example of this race
    example = next(ex for ex in train_dataset if ex['race'] == race)

    axes[idx].imshow(example['image'])
    axes[idx].set_title(f"{race}")
    axes[idx].axis('off')

plt.tight_layout()
plt.show()
```



How did they answer their questions?

2 Algorithms

- Face recognition
- Gender assignment

There are MANY options for feature extraction, such as...

- ▶ **Color Histograms:** Represents the distribution of color values in an image.
- ▶ **Texture Features:** Calculate texture features like entropy, energy, contrast, and homogeneity from the grayscale co-occurrence matrix.
- ▶ **Histogram of Oriented Gradients (HOG):** Compute the histogram of gradient orientations in localized portions of the image.
- ▶ **Haar-like Features:** Use integral images to efficiently compute features that recognize edges, lines, and rectangles for object detection.

→ Different feature types provide different kinds of information about the image content, and the best one to use depends on the specific task at hand!

How did they answer their questions?

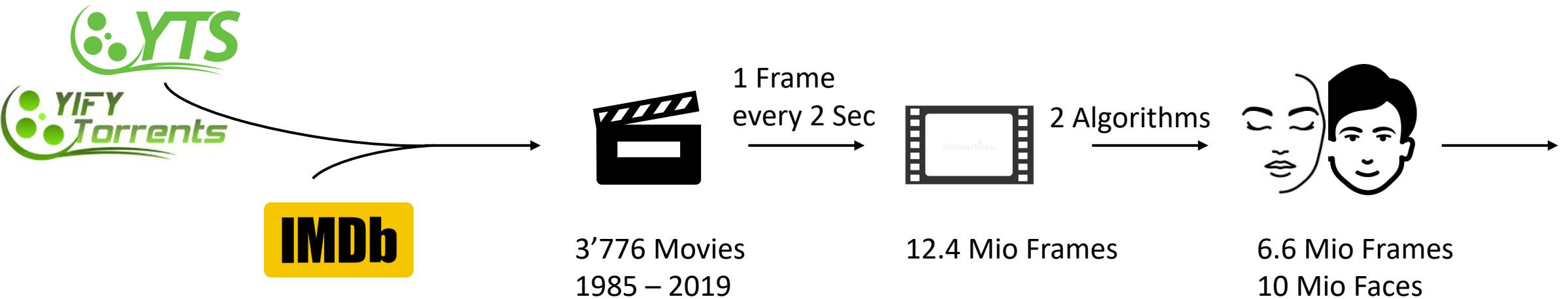
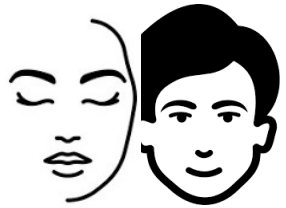


Fig 1 Antoine Mazières et al 2021

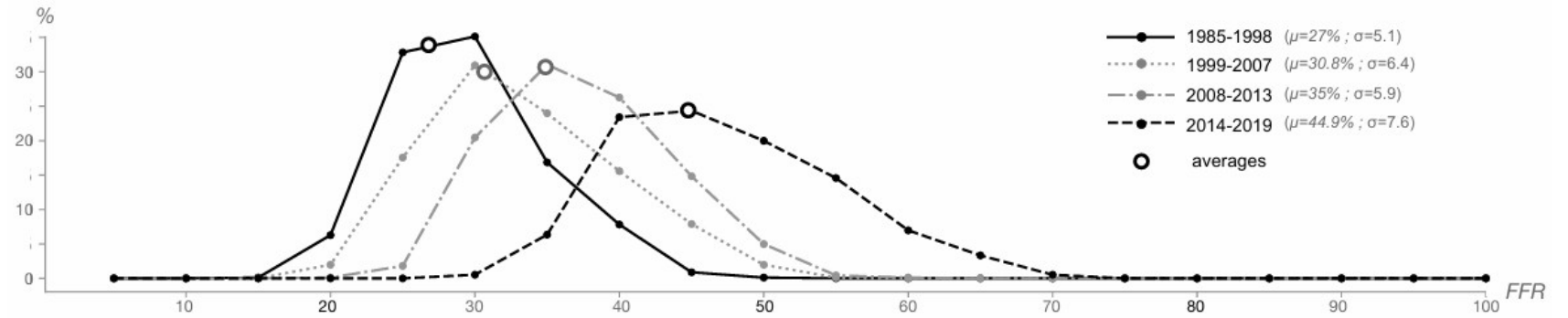
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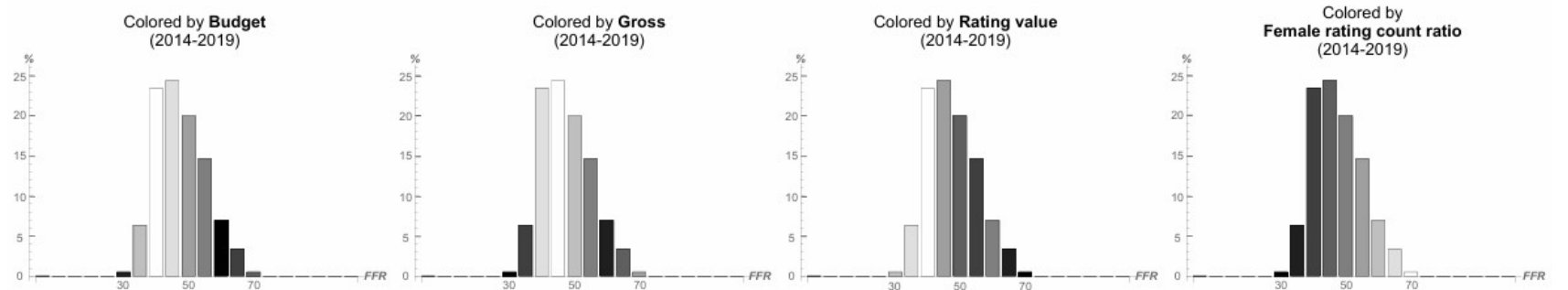
What did they find? – It is getting better, but....



6.6 Mio Frames
10 Mio Faces



(a) Distributions of FFR for each period.



(b) Distributions of several features over the distribution of FFR.

What did they find?

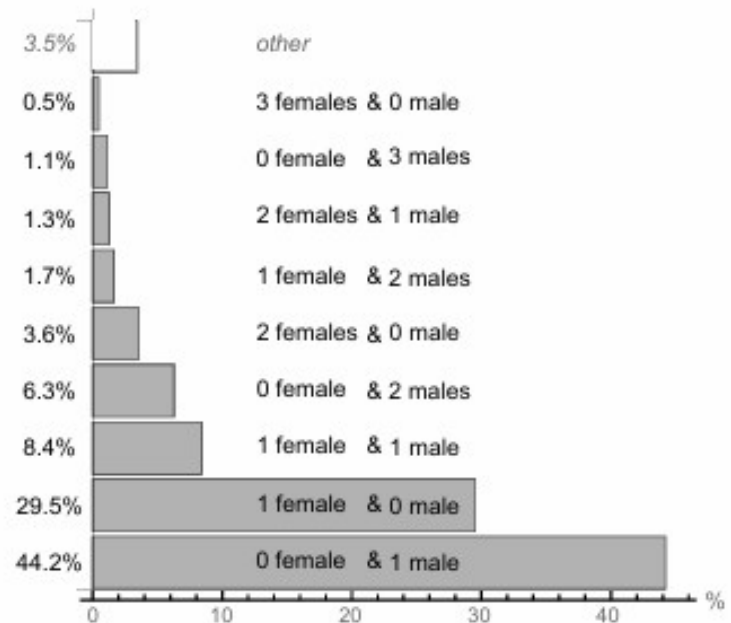


Figure 5: **Combinations of character gender** (2014-2019).

Conclusion

- How under- or over-represented are women on-screen in popular movies?
 - It is **getting better** but women are still under-represented
- Can computational methods (face and gender detection) measure this representation more efficiently and accurately over time?
 - Yes, it is possible to answer the question above with computational methods and their accuracy is close to literature values

Conclusion – convincing?

- Yes:
 - Large, diverse dataset and transparent **validation**
 - Quantitative **correction** for algorithmic bias
- Potential for **computational social science** in gender studies
- Highlights need for **algorithmic fairness** and transparency
- Policy relevance: ongoing **gender imbalance**, but **progress** toward parity
- Informs about **diversity targets** in media production

My Questions for you

- Which option would you choose for the feature extraction in the two algorithms, based on the options discussed in the 2nd lecture?
 - Do you see any bias in the data acquisition?
 - Should we unveil and tackle gender inequalities by using ML and do you think it has an impact?
- ▶ **Color Histograms:** Represents the distribution of color values in an image.
 - ▶ **Texture Features:** Calculate texture features like entropy, energy, contrast, and homogeneity from the grayscale co-occurrence matrix.
 - ▶ **Histogram of Oriented Gradients (HOG):** Compute the histogram of gradient orientations in localized portions of the image.
 - ▶ **Haar-like Features:** Use integral images to efficiently compute features that recognize edges, lines, and rectangles for object detection.

Thank you – Questions?